

Roll No.

TCS–252

B. Tech. (Fourth Semester)

EXAMINATION, 2024-25

DESIGN ANALYSIS OF ALGORITHM

Time : $2\frac{1}{2}$ Hours

Maximum Marks : 60

Note : Question paper has three Sections. Read instructions carefully for each Section.

Section—A

(Very Short Answer Type Questions)

1. Attempt all questions. 1 each
 - (a) Name the different techniques used for designing algorithms. (CO1, RE)
 - (b) Solve the recurrence relation : (CO1, AP)

$$T(n) = 4T\left(\frac{n}{2}\right) + n.$$

P. T. O.

- (c) Compare the time efficiency of Insertion Sort or Merge sort. (CO2, UN)
- (d) Write the time complexity of the Longest Common Subsequence algorithm. (CO2, RE)
- (e) Define the Traveling salesman problem with an example. (CO2, UN)
- (f) Give two examples of the Greedy approach. (RE)
- (g) Explain the divide and conquer strategy with an example. (CO2, RE)
- (h) algorithm is suitable for +ve weighted graph to solve the single source shortest path. (CO4, UN)
- (i) algorithms are used to solve NP problems. (CO5, UN)
- (j) If there are 65 keys in a binomial heap, then the structure consists of binomial trees. 1(UN)
- (k) Prove that the statement is correct or incorrect, $f(n) = 3n + 5 \in o(n^2)$. (CO1, AN)
- (l) Find the big-oh notation for $f(n) = 5 \cdot 2^n + 3n + 5$. (CO1, EV)

Section—B**(Short Answer Type Questions)**

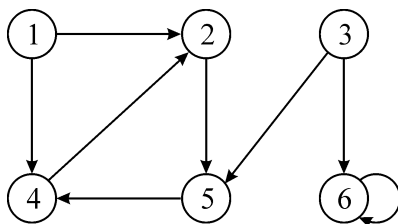
2. Attempt all questions.

- (a) Define the classes problems P, NP and NPC. How are they related to each other ? Is every decision problem an NP couple ? 4(CO5, RE)

OR

Show the comparisons the naïve string matcher makes for the pattern $P = 10001$ in the text $T = 0000100010010$. (CO5, EV)

- (b) Write an algorithm for DFS and implement it on the below graph (figure-1). 4(CO4,AP)

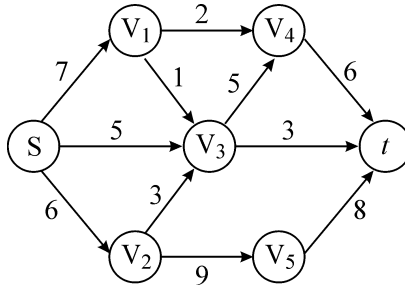


OR

In the flow network illustrated in Figure-2, each directed edge is labeled with its capacity.

P. T. O.

Use the Ford-Fulkerson algorithm to find the maximum flow in the network. (CO4, AP)



- (c) Show the red-black tree that results after successful inserting the keys 140, 137, 130, 111, 118, 17 into an initially empty red-black tree. 4(CO3, AP)

OR

Write the properties of B-Tree. Prove that a red-black tree with n internal nodes has a height of at most $2\lg(n + 1)$. (CO3, EV)

- (d) Find the optimal solution for the 0/1 knapsack problem using DPP, given that the capacity of the knapsack is 50, $\text{val[]} = \{10, 40, 30, 50\}$, $\text{wt[]} = \{50, 40, 60, 30\}$ 4(CO2, AP)

OR

Explain the Backtracking algorithm and generate the result for the 4-queens problem.

(CO2, AP)

- (e) Solve the recurrence relation : (CO1, AP)

$$T(n) = T\left(\frac{2n}{3}\right) + n.$$

OR

Solve the recurrence relation : (CO1, AP)

$$a_n - 4a_{n-1} + 4a_{n-2} = 3^n + 4.$$

- (f) Define the term algorithm. What are the various features of an algorithm ?

4(CO1, UN)

OR

Let $T_1(n)$ & $T_2(n)$ are time complexities of two program segments P1 & P2, where $T_1(n) \in O\{f(n)\}$ and $T_2(n) \in O\{g(n)\}$. Then, what is the complexity of a program fragment P1 followed by P2? (CO1, UN)

P. T. O.

Section—C**(Long Answer Type Questions)**

3. Attempt all questions.

- (a) Write an algorithm for Linear Search and derive its complexity. 8(CO1, UN)

OR

Write an algorithm for binary Search and derive its complexity. (CO1, UN)

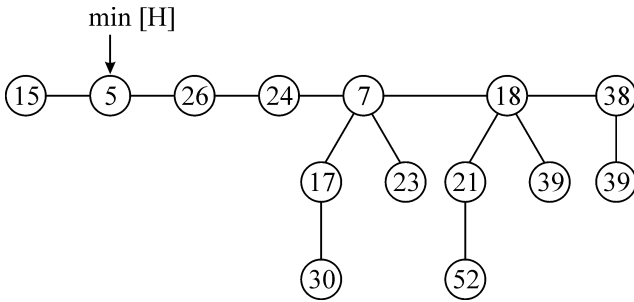
- (b) Let $A = \langle 7, 2, 4, 17, 1, 11, 6, 8, 15, 20, 20 \rangle$

8(CO3, AP)

- (i) Form a binomial heap whose keys are elements of A.
- (ii) From the binomial heap obtained, apply the operation of extracting the node with the minimum key.

OR

Show the steps to extract the min-key from the given Fibonacci heap (Figure 3)(CO3, AP)

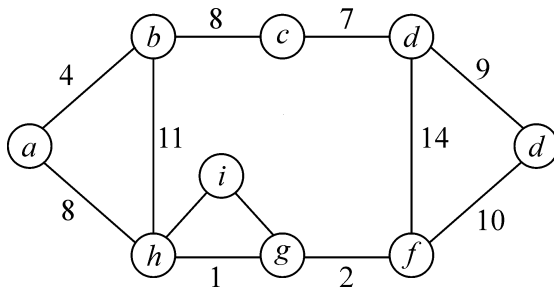


- (c) Find an optimal parenthesization of a matrix chain product whose sequence of dimension is {4, 10, 3, 12, 20, 7}. 8(CO2, AP)

OR

Using Kruskal's algorithm, generated the minimum cost spanning tree for the following Figure 4. Courdeu vertex @ as source vertex.

(CO2, AP)



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